

**Project Proposal  
for  
Development of  
Model Village Mission & Mahapurusha  
Adarsha Gram Yojana – Monitoring &  
Management Portal**

**VIKSHIT ODISHA VISION 2036 & 2047**

Submitted to:  
**Panchayati Raj & Drinking Water Department  
Government of Odisha**

Prepared by:



**Odisha State Centre  
National Informatics Centre  
Ministry of Electronics & Information Technology  
Government of India**

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## **Executive Summary**

The Government of Odisha, under the Viksit Odisha Vision 2036 & 2047, has envisioned holistic and sustainable rural development through targeted interventions, inter-departmental convergence and evidence-based planning. As part of this vision, two flagship initiatives—Model Village Mission (MVM) and Mahapurusha Adarsha Grama Yojana (MAGY)—aim to transform villages and Gram Panchayats into model rural settlements by addressing critical developmental gaps across multiple sectors.

The Model Village Mission focuses on assessing and improving the developmental status of villages through comprehensive surveys covering multiple departments, while Mahapurusha Adarsha Grama Yojana (MAGY) aims to recognize and develop the native Gram Panchayats of eminent personalities from Odisha by transforming them into model Gram Panchayats that reflect holistic rural development. In the initial phase, the proposed system will cover approximately 6,733 villages under the Model Village Mission and 16 Gram Panchayats under MAGY, with the platform designed to subsequently scale across all villages and Gram Panchayats in the State.

NIC, Odisha State Centre, proposes to design, develop and deploy a centralized, web-based Monitoring & Management Portal for the Model Village Mission and Mahapurusha Adarsha Grama Yojana (MAGY). The portal will digitize the entire survey-to-decision lifecycle—from survey data capture at the Block level for the Model Village Mission and at the Gram Panchayat level for MAGY, through multi-level validation, District-level certification, State-level review, and controlled access for the respective Departmental Nodal Officers. The solution will support survey management for 11 participating departments under the Model Village Mission and 18 departments under MAGY. Developed using a modern and scalable technology stack comprising React.js (frontend), Node.js (backend) and PostgreSQL (database), the portal will provide centralized data management, real-time KPI dashboards, developmental gap analysis, inter-departmental convergence, DPR preparation and management, document and evidence management, audit trails, and continuous progress monitoring with role-based workflows.

This document comprises the study, design, development and deliverables regarding our understanding of user requirements, technical solutions, delivery time frame and terms & conditions. It provides the user Department with an outline of our work procedures & assumptions of the project. It also specifies the scope, cost estimate and functionalities for development of Model Village Mission & Mahapurusha Adarsha Gram Yojana Monitoring & Management Portal for Panchayati Raj & Drinking Water Department, Government of Odisha.

Once implemented, the portal will serve as the single source of truth for village-level development data in Odisha, enabling transparent, timely, and evidence-based planning in line with the Vikshit Odisha 2036 & 2047 goals.

The estimated project cost of Rs. xxxx /-(Rupees xxxx only) is for the first one year for development/ implementation/ maintenance/ security audit/ SSL certificate. If the user decides to host the application in NDC, NIC then cost estimate of hosting charge will be furnished separately. If the application will be hosted in SDC, PR & DW Department, Government of Odisha will have to contact E & IT Department, Government of Odisha. Cost towards SMS service charges as applicable will be borne by the user department additionally. If there is any change/addition in requirement over and above the scope of work given, the cost estimate will be revised accordingly.

## Approval Matrix

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## Version Control

| Version | Date          | Brief Description        | Section Change                                                                                   |
|---------|---------------|--------------------------|--------------------------------------------------------------------------------------------------|
| 1.0     | xx-July -2026 | Project Proposal created | To design, develop and implementation of Model Village Mission – Monitoring & Management Portal. |

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## **1. Introduction**

The Panchayati Raj & Drinking Water Department, Government of Odisha, plays a pivotal role in strengthening rural governance through the three-tier Panchayati Raj Institutions (PRIs) and in implementing major Central and State-sponsored programmes aimed at rural development, infrastructure creation, poverty alleviation and improvement of the quality of life in rural areas. The Department has been instrumental in driving Odisha's rural transformation by promoting decentralized planning, effective service delivery and sustainable development through active community participation.

As part of the Viksit Odisha Vision 2036 & 2047, the Department is implementing flagship initiatives such as the Model Village Mission (MVM) and the Mahapurusha Adarsha Grama Yojana (MAGY) to accelerate holistic rural development. While the Model Village Mission focuses on transforming villages through targeted interventions based on identified developmental gaps, Mahapurusha Adarsha Grama Yojana (MAGY) aims to recognize, commemorate and develop the native Gram Panchayats of eminent personalities from Odisha who have made significant contributions to the State's social, cultural, literary and political history, by transforming them into model Gram Panchayats that reflect holistic and sustainable rural development.

To support these initiatives, comprehensive baseline surveys are being conducted to capture village and Gram Panchayat-level developmental indicators across multiple sectors. The surveys cover 11 participating departments under the Model Village Mission and 18 participating departments under MAGY, including Panchayati Raj & Drinking Water, Health & Family Welfare, Mission Shakti and other line departments. The questionnaires comprise structured indicators that assess the availability of infrastructure, public services and other developmental parameters, providing a comprehensive baseline for planning, implementation and monitoring of development initiatives.

Considering the scale of these programmes and the involvement of multiple stakeholders across Gram Panchayat, Block, District and State levels, there is a need for a centralized digital platform to streamline survey data collection, validation of survey data and monitoring. The proposed workflow facilitates survey data entry at the Gram Panchayat level

for MAGY and at the Block level for the Model Village Mission, followed by multi-level validation, certification by the District Collector, State-level review and controlled access for the respective Departmental Nodal Officers.

Beyond baseline data collection, the proposed portal will also serve as a continuous monitoring and management platform. Developmental indicators initially marked as unavailable ("No") during the baseline survey can subsequently be updated by authorized officials upon completion of the corresponding developmental activities, along with supporting evidence such as photographs and relevant documents. Each update will undergo the prescribed validation and approval workflow, enabling continuous monitoring of progress until all identified developmental gaps are addressed. The portal will further provide real-time dashboards, KPI monitoring, gap analysis, inter-departmental convergence, project monitoring and comprehensive MIS reports to facilitate evidence-based decision-making and transparent governance.

Given the large scale of the initiative and the involvement of multiple administrative levels and departments, a centralized digital platform is essential for the effective collection, validation, management and utilization of this information. The data needs to move through a structured workflow involving data entry, Block-level validation, District-level validation, certification by the District Collector, State-level review and subsequent access by concerned Departmental Nodal Officers.

The information in this document is not being disclosed except for the evaluation of initial analysis by the recipient's employees or representative.

## **2. Scope of the Project**

The scope of work covers the design, development, implementation, deployment and maintenance of a comprehensive Monitoring and Management Portal for the Model Village Mission (MVM) and Mahapurusha Adarsha Grama Yojana (MAGY) under the Viksit Odisha Vision 2036 & 2047. The proposed portal will provide a unified digital platform for capturing, validating, certifying, analysing and continuously monitoring developmental indicators, enabling evidence-based planning, inter-departmental convergence and tracking of rural development initiatives.

The portal will initially cover approximately 6,733 villages under the Model Village Mission (MVM) and 16 Gram Panchayats under the Mahapurusha Adarsha Grama Yojana (MAGY) during Phase-I implementation. Subsequently, the Model Village Mission will be expanded to cover approximately 50,000 villages across Odisha, while the coverage under MAGY will be extended to approximately 100–150 Gram Panchayats. The proposed solution will be developed on a scalable and configurable architecture to support the phased implementation and future enhancement of both schemes.

1. Master Data Management: Management of districts, blocks, Gram Panchayats, villages, departments, sectors and developmental indicators.
2. Questionnaire Management: Creation and management of dynamic, department-wise and sector-wise questionnaires, including Yes/No, multiple-choice, numeric and other structured questions.
3. Survey Data Management: Village-level data entry, draft saving, submission, editing and supporting document upload.
4. Multi-Level Validation Workflow: Implementation of the defined workflow from Block-level validation to District-level validation, Collector certification, State-level review and release to Departmental Nodal Officers.
5. Role-Based Access Control: Access management based on user role, department and administrative jurisdiction.
6. Village Developmental Profile: Consolidated village-wise view of developmental indicators and sector-wise status.
7. KPI Monitoring and Gap Analysis: Monitoring of key indicators and identification of village-wise, sector-wise and department-wise developmental gaps.
8. Inter-Departmental Convergence: Identification and coordination of interventions required from multiple departments for holistic village development.
9. DPR and Project Management: Preparation, submission, review and monitoring of Detailed Project Reports and associated developmental projects.
10. MIS and Dashboards: Development of role-based dashboards and reports for State, District, Block and Departmental authorities.

11. Document and Evidence Management: Secure management of survey evidence, DPRs, progress documents and other supporting files.
12. Audit Trail and Monitoring: Tracking of data entry, validation, modification, certification and other user activities.
13. Security and Scalability: Implementation of appropriate security, backup, monitoring and scalable architecture to support the initial onboarding of approximately 6,733 villages and subsequent expansion to approximately 50,000+ villages across Odisha.
- 14. Alerts & Notifications**
  - a. Auto-notifications for deadlines, non-compliance, overdue replies, and updates via email/SMS

## **Objective of the Project :**

The proposed Monitoring & Management Portal for the Model Village Mission (MVM) and Mahapurusha Adarsha Grama Yojana (MAGY) aims to achieve the following objectives:

1. Digitize the collection and management of village- and Gram Panchayat-level developmental data through configurable, scheme-specific and department-wise questionnaires.
2. Establish a secure, multi-level workflow for survey submission, validation, certification and review involving Gram Panchayat, Block, District, Collector and State-level authorities.
3. Provide role-based, secure access to all stakeholders, including GPEOs/ADEOs, Additional BDOs, CDOs, District Collectors, State PR & DW Department officials and Departmental Nodal Officers.
4. Create a centralized repository of validated developmental data to support planning, implementation and monitoring of rural development initiatives.
5. Enable real-time dashboards, MIS reports and Key Performance Indicator (KPI) monitoring to assess developmental status, identify gaps and track the progress of villages and Gram Panchayats.
6. Facilitate inter-departmental convergence by providing a unified platform for sharing developmental information and coordinating interventions across participating departments.

7. Support the preparation, submission and monitoring of Detailed Project Reports (DPRs) and developmental projects based on validated survey findings.
8. Enable continuous progress monitoring by allowing authorized officials to update developmental indicators initially marked as "No" upon completion of the corresponding work, supported by documentary evidence, with each update undergoing the prescribed validation workflow until all identified developmental gaps are addressed.
9. Maintain a comprehensive audit trail of survey submissions, validations, approvals, progress updates, evidence uploads and all other user activities to ensure transparency, accountability and good governance.
10. Develop a secure, scalable and configurable platform capable of supporting phased implementation of both schemes, onboarding additional departments, indicators and future enhancements in alignment with the objectives of Viksit Odisha Vision 2036 & 2047.

### **3. Organisation Description (NIC)**

National Informatics Centre (NIC) is a premier Science and Technology Organization under the Ministry of Electronics and Information Technology, Government of India actively working for the past four decades in the area of Information and Communication Technology (ICT) Applications in the Government Sector. It has been providing network backbone and e-Governance support to Central Government, State Governments, UT Administrations, Districts and other Government bodies. It offers a wide range of ICT services including Nationwide Communication Network for decentralized planning, improvement in Government services and wider transparency of national and local Governments. NIC assists in implementing Information Technology Projects, in close collaboration with Central and State Governments, in the areas of (a) Centrally sponsored schemes and Central sector schemes, (b) State sector and State sponsored projects, and (c) District Administration sponsored projects. NIC endeavours to ensure that the latest technology in all areas of IT is available to its users.

#### **National Informatics Centre Service Incorporated (NICSI)**

National Informatics Centre Services Inc. (NICSI) was established in 1995 as a section-25 (now section 8 company under the Companies Act, 2013) company under National Informatics Centre, Ministry of Electronics & Information Technology, Government of India for providing and procuring IT solutions for multiple e-governance projects undertaken by NIC, MeitY, Governments and Government Organizations (like Public Sector Undertakings).

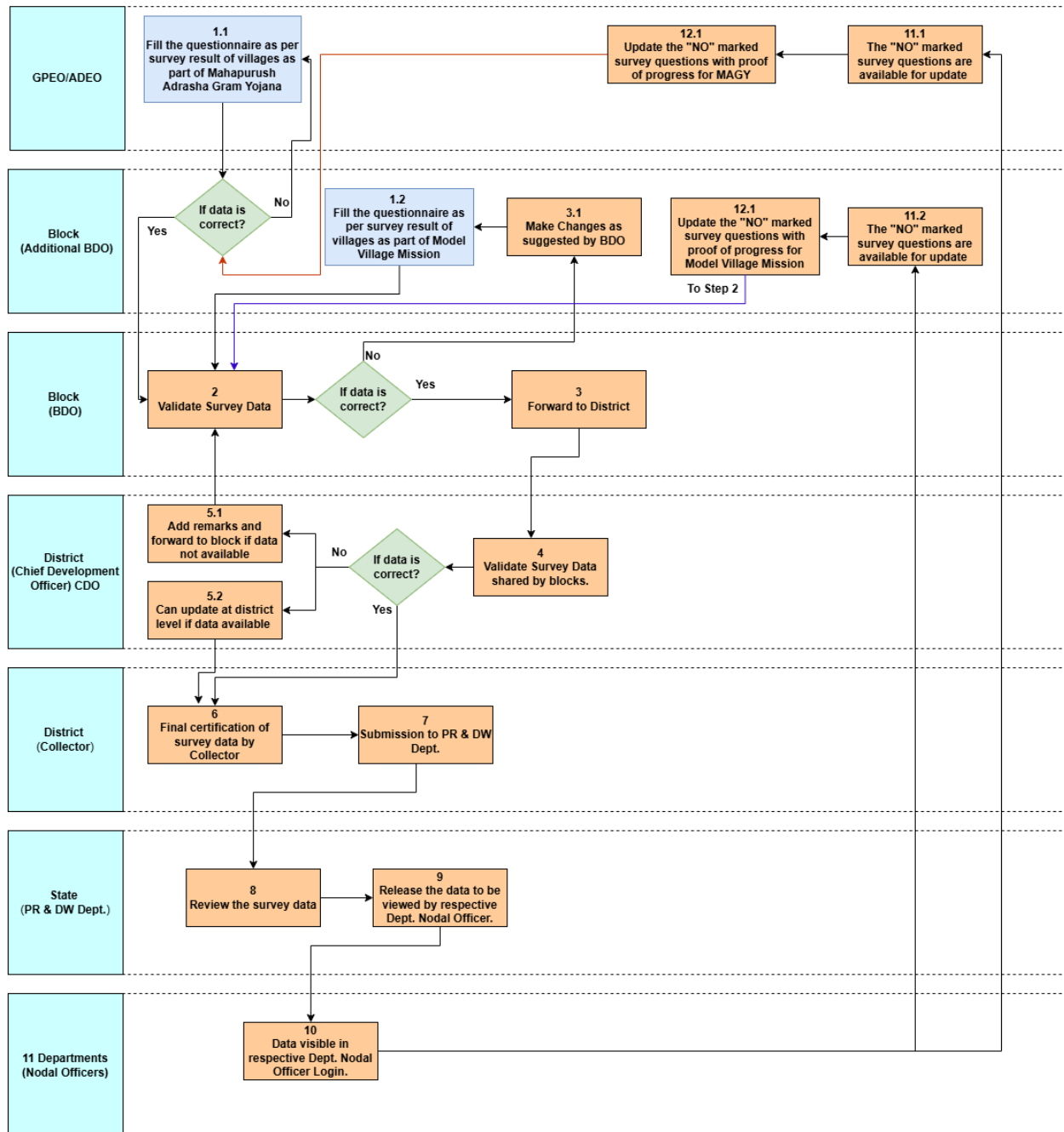
## **4. Proposed System**

The proposed Monitoring & Management Portal will provide a unified digital platform for implementing and monitoring both the Model Village Mission (MVM) and the Mahapurusha Adarsha Grama Yojana (MAGY) under Viksit Odisha Vision 2036 & 2047.

The portal will facilitate the complete lifecycle of village and Gram Panchayat development—from baseline survey data collection and multi-level validation to developmental gap analysis, project planning, continuous progress monitoring and performance evaluation. The workflow has been designed to ensure data accuracy, accountability and seamless coordination among stakeholders at Gram Panchayat, Block, District and State levels.

The overall workflow of the proposed portal is illustrated in the flowchart below.

### Model Village Mission & Mahapurusha Adarsha Gram Yojana Monitoring & Management Portal



## 4.1 Workflow

### 1. Survey Data Entry

Survey data shall be captured based on approved field survey reports using structured, department-wise questionnaires.

Model Village Mission (MVM): Survey data shall be entered by the designated Assistant Engineer/Additional BDO at the Block level for approximately 6,733 villages during Phase-I.

Mahapurusha Adarsha Grama Yojana (MAGY): Survey data shall be entered by the designated Gram Panchayat Extension Officer (GPEO)/Assistant Development Extension Officer (ADEO) at the Gram Panchayat level for 16 Gram Panchayats during the initial phase, with subsequent expansion to 100–110 Gram Panchayats.

The questionnaires shall capture department-specific developmental indicators comprising predominantly Yes/No responses along with other structured data elements.

## **2. Block-Level Validation**

The submitted survey data shall be reviewed by the designated Block-level authority for validation.

If discrepancies are identified, the record may be returned to the originating official with remarks for correction. After necessary modifications, the survey shall be resubmitted for validation.

Upon successful validation, the record shall be forwarded to the District level.

## **3. District-Level Validation**

The Chief Development Officer (CDO) shall verify the validated survey data.

The CDO may:

- Validate the submitted data.
- Return the data with remarks for correction.
- Update verified information available at the District level, wherever authorized.

Corrected records shall be resubmitted through the prescribed workflow until validation is completed.

## **4. District-Level Certification**

After successful validation, the District Collector shall perform the final certification of the survey data for onward submission to the State Government.

## **5. State-Level Review**

The Panchayati Raj & Drinking Water Department shall review the certified survey data and make it available to the respective Departmental Nodal Officers for planning and monitoring purposes.

## **6. Departmental Access**

The released data shall be accessible to the respective Departmental Nodal Officers.

- Model Village Mission: Data shall be available to 11 participating departments.
- Mahapurusha Adarsha Grama Yojana (MAGY): Data shall be available to 18 participating departments.

Departmental users shall be able to analyse developmental indicators, identify gaps, generate reports and monitor scheme-specific progress.

## **7. Continuous Progress Monitoring**

Following the baseline survey and departmental review, the portal shall facilitate continuous monitoring of developmental progress.

Developmental indicators initially marked as "No" may subsequently be updated by authorized officials after completion of the corresponding developmental activity.

- For Model Village Mission: Progress updates shall be submitted by the designated Block-level officials.
- For MAGY: Progress updates shall be submitted by the designated GPEO/ADEO at the Gram Panchayat level.

Whenever an indicator is updated from "No" to "Yes", the user shall upload supporting evidence, such as photographs or relevant documents.

Every progress update shall follow the same validation workflow comprising:

Progress Update → Block Validation → District Validation → Collector Certification  
→ State Review → Departmental Dashboard

This cycle shall continue until all identified developmental gaps across participating departments are addressed.

## **8. Monitoring, MIS and Dashboards**

The validated data shall be used to generate real-time dashboards and analytical reports for monitoring the implementation of both schemes.

The portal shall provide:

- Scheme-wise progress monitoring
- Department-wise performance
- Village and Gram Panchayat development status
- KPI monitoring
- Gap analysis
- Completion percentage
- Pending developmental indicators
- District-wise and Block-wise progress
- Evidence verification status
- Project and DPR monitoring

The validation and correction mechanism throughout the workflow shall ensure progressive improvement of data quality while maintaining a comprehensive audit trail of every submission, correction, validation, certification and progress update.

## 4.2 Functional Modules

| Module                            | Key Capabilities                                                                                                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. User & Role Management         | Role-Based Access Control (RBAC) for GPEO/ADEO, Assistant Engineer/Additional BDO, CDO, District Collector, State PR & DW Department, Departmental Nodal Officers and System Administrator, with jurisdiction-based access. |
| 2. Questionnaire Configuration    | Admin-configurable, department-wise, dynamic questionnaire builder (Yes/No) without code changes.                                                                                                                           |
| 3. Survey Data Entry              | Scheme-specific survey data entry, draft save, validation rules, document upload and submission workflow.                                                                                                                   |
| 4. Validation & Remarks Workflow  | Multi-level validation with review queues, remarks, correction workflow, resubmission tracking and configurable approval process.                                                                                           |
| 5. Certification Workflow         | Collector-level digital certification/sign-off with approval trail.                                                                                                                                                         |
| 6. State Review & Release         | State-level review, approval and controlled release of validated data to Departmental Nodal Officers.                                                                                                                       |
| 7. Departmental Portal            | Department-specific access to survey data, developmental indicators, reports and dashboards for participating departments.                                                                                                  |
| 8. Continuous Progress Monitoring | Updating of pending ("No") developmental indicators with supporting evidence, re-validation workflow, progress tracking and historical record management.                                                                   |
| 9. MIS & Dashboards               | Real-time dashboards, development scorecards, gap analysis, progress monitoring, completion tracking and drill-down analytical reports.                                                                                     |
| 10. DPR Support                   | Export of consolidated, department/village-wise data in report-ready formats to support DPR preparation and submission.                                                                                                     |
| 11. Notifications & Alerts        | Email/SMS/in-app alerts for pending validations, remarks raised, certification due, data released.                                                                                                                          |
| 12. Audit Trail & Logs            | Immutable log of every entry, edit, remark, and approval with user, timestamp, and IP for governance and grievance redressal.                                                                                               |
| 13. Admin & Master Data           | Village/block/district/GP master data management, department master, indicator master, mission/phase configuration.                                                                                                         |

## **4.3 Deliverables**

Following deliverables will be generated from the application/project,

1. Alert/Notification through SMS/e-Mail on
2. Dashboard on KPI
3. MIS reports
4. Analytical Report

## **4.4 Proposed Technology**

The technologies to be used for the development of this application software are given below.

### **Web Application**

Server: Linux -Ubuntu-22, Apache/Nginx, Tomcat

Database: Postgresql (Open source) /MongoDB

Backend: Java-Spring Boot/ NodeJS

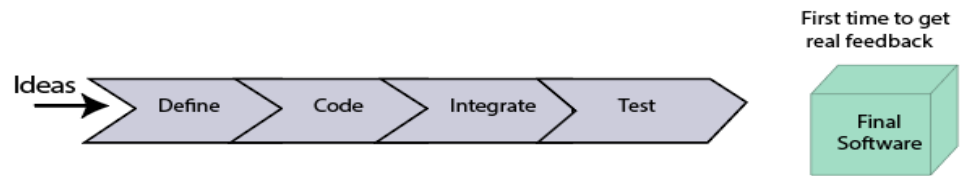
Frontend: Angular/React

**Deployment Environment:** Cloud

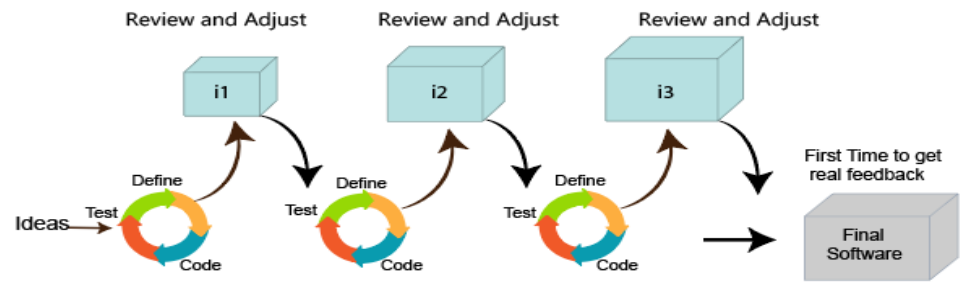
## **4.5 Proposed Approach and Methodology**

We will follow SCRUM(agile development methodology) for this proposed project.

- An agile methodology is an iterative approach to software development. Each iteration of agile methodology takes a short time interval of 1 to 4 weeks. The agile development process is aligned to deliver the changing business requirement. It distributes the software with faster and fewer changes.
- The single-phase software development takes 6 to 18 months. In single-phase development, all the requirement gathering and risks management factors are predicted initially.
- The agile software development process frequently takes the feedback of workable product. The workable product is delivered within 1 to 4 weeks of iteration.



Traditional Method



Agile Method

## 4.6 Design and Development Principle

### RESTful API based architecture

The architecture of Dashboard Monitoring System will be based on RESTful APIs which will structure the platform as a set of loosely coupled, collaborating services. A service typically will be a set of distinct features or, functionality. It will support a variety of different clients e.g. desktop browsers, mobile browsers and native mobile applications. The services would communicate directly and synchronously with each other over HTTP/HTTPS, without the need for any additional infrastructure.

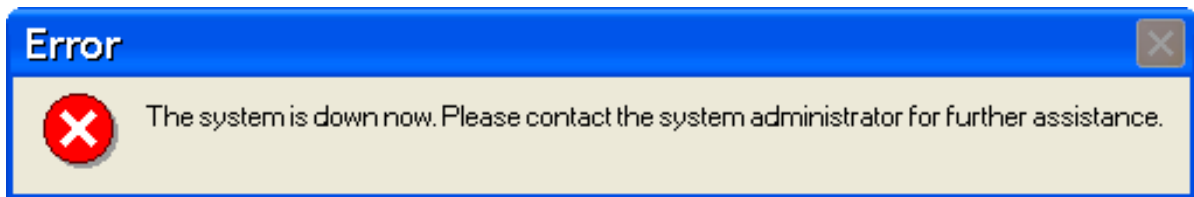
### Data Access Object

The Data Access Object (DAO) design principle is mechanism to abstract away the details of persistence in an application. The idea is that instead of having the domain logic communicate directly with the database, the business logic speaks to a DAO layer instead. This DAO layer then communicates with the underlying persistence system or service.

### Error Management

The proper error management approach would be implemented to handle errors in VE-IT platform. There would be mainly two types of error that would be handled –

**Technical** – These errors would be mainly related to malfunction of application. E.g. - any error when trying to load a page, JavaScript error, the application hangs for a long time, the application crashes, etc.



All the code would be written within try-catch block. The layer that will face an exception, raise a new exception to the upper layer if needed. The final layer will decide the proper message to show to the users and redirect the user to a special page that will contain the proper message. An email notification will be sent to system administrator with the detail of error.

**Functional** – These errors would be raised when any business validation or, business rules would fail. This would be further clarified during development process as per the need and as per the module.

#### 4.7 Cyber Security Auditing

For cyber security, OWASP top ten 2021 will be complied the web application after getting the UAT.

#### 4.8 Training

NIC shall provide 6 days training to user/administration and train them on the overall workflow of the developed solution and backend administration process. User manuals covering the detail technical description of the software solution, implementation and operational procedures will be provided. Logistic support for the training is to be provided by PR & DW Department.

#### 4.9 Implementation Plan

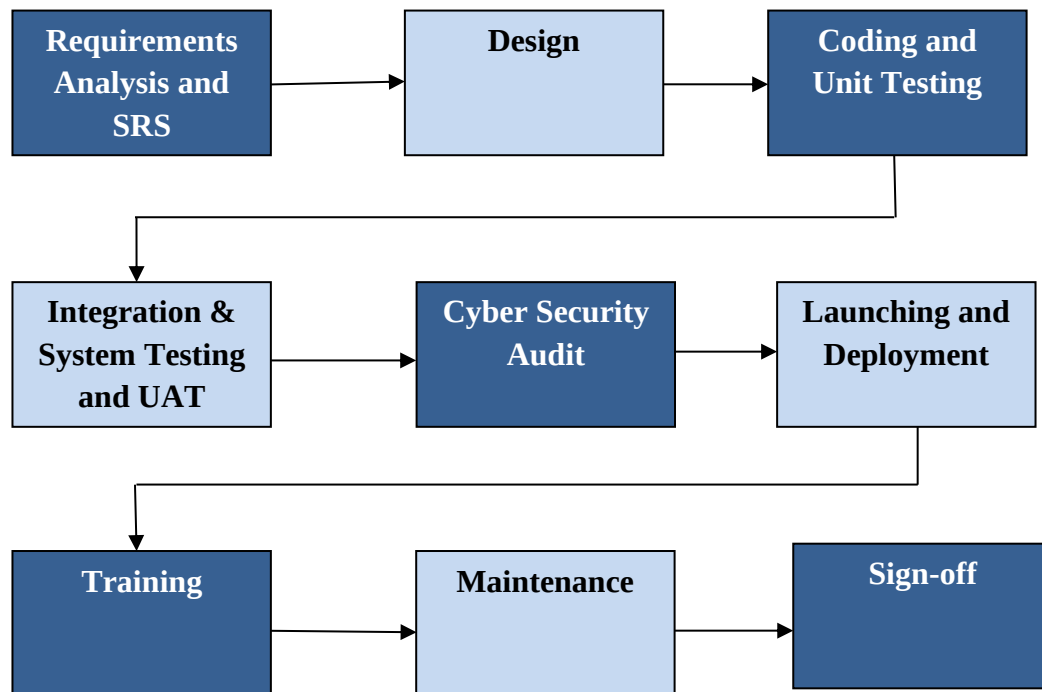
Hosting of the application will be in cloud environment, as described in this document in x.x below.

After getting the UAT, the application will be deployed for pilot implementation for PR & DW Department, one district level office, one block level office and one GP level office for field level/end-user level feedback and functionalities testing. After getting issues/feedback from the pilot users if any, will be fixed within 2 weeks from the day of receipt. Thereafter the application will be ready for implementation in all over Odisha.

## 5. Estimated Project Timeline

| Sl. No. | Task No. | Activity                                                                                                                    | Timeline                      | Concerned Organisation | Remarks |
|---------|----------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------|---------|
| 1       | Task 0   | Requisition letter from PR & DW Dept and Primary study of the system, meeting/discussion with the officials of PR & DW Dept |                               | PR & DW Dept.          |         |
| 2       | Task 1   | Submission of Project Proposal                                                                                              | 15 days of Task 0             | NIC                    |         |
| 3       | Task 2   | Approval of PP & Release of funds                                                                                           | 15 days                       | PR & DW Dept.          |         |
| 4       |          | Project Team Formation                                                                                                      |                               |                        |         |
| 4.1     |          | NIC Project Team                                                                                                            |                               | NIC                    |         |
| 4.2     |          | Project Review Committee                                                                                                    |                               | PR & DW Dept.          |         |
| 5       | Task 3   | Submission of SRS (Software Requirements Specification) Document                                                            | 21 days from Release of funds | NIC                    |         |
| 6       | Task 4   | Approval of SRS                                                                                                             | 15 days                       | PR & DW Dept.          |         |
| 7       | Task 5   | Design                                                                                                                      | 30 days from approval of SRS  | NIC                    |         |
| 8       | Task 6   | Coding/Development, Unit Testing, Integration & System Testing                                                              | 45 days                       | NIC                    |         |
| 9       | Task 7   | User Acceptance Testing                                                                                                     | 7 days                        | PR & DW Dept.          |         |
| 10      | Task 8   | Cyber Security Audit                                                                                                        | 45 days                       | NIC                    |         |
| 11      | Task 9   | Launching , Deployment and Training                                                                                         | 15 days                       | NIC                    |         |
| 17      | Task 10  | Compliance of user feedbacks                                                                                                | Rest days upto 1 year         | PR & DW Dept. and NIC  |         |

## Project Progress Plan (for 1 year):



## Resource Plan

| Sl. No. | Activity                                           | Manpower Type                                                          | Number of Resources | Number of Months | Location |
|---------|----------------------------------------------------|------------------------------------------------------------------------|---------------------|------------------|----------|
| 1       | 2                                                  | 3                                                                      | 4                   | 5                | 6        |
| 1       | Software Development, Implementation & Maintenance | Software Application Support Engineer (Level 4 with 1 year experience) |                     |                  | NIC      |
| 2       | Software Development, Implementation & Maintenance | Software Application Support Engineer (Level 3 with 1 year experience) |                     |                  | NIC      |

## 6. Estimated Project Cost

### Application Design/Development/Implementation

| Sl. No. | Activity                                                              | Manpower Type                                                          | Minimum Monthly Remuneration (INR) | Operating Margin (OM : 4.94%) | NICSI Margin @ 9% of (column4+column5) | GST @ 18% of (column4+column5 +column6) | Unit Price in INR inclusive all | Number | Man Month | Estimated Cost in INR (column8 X column9 X column10)                   |
|---------|-----------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------|-------------------------------|----------------------------------------|-----------------------------------------|---------------------------------|--------|-----------|------------------------------------------------------------------------|
| 1       | 2                                                                     | 3                                                                      | 4                                  | 5                             | 6                                      | 7                                       | 8                               | 9      | 10        | 11                                                                     |
| A       | Software Development, Implementation & Maintenance                    | Software Application Support Engineer (Level 4 with 1 year experience) |                                    |                               |                                        |                                         |                                 |        |           |                                                                        |
| B       | Software Development, Implementation & Maintenance                    | Software Application Support Engineer (Level 3 with 1 year experience) |                                    |                               |                                        |                                         |                                 |        |           |                                                                        |
| C       | Security Audit of Web Application and Mobile Application ( one time ) |                                                                        |                                    |                               |                                        |                                         |                                 |        |           |                                                                        |
| D       | SSL Certificate ( for 02 years)                                       |                                                                        |                                    |                               |                                        |                                         |                                 |        |           |                                                                        |
| E       | Travel & Communication                                                |                                                                        |                                    |                               |                                        |                                         |                                 |        |           |                                                                        |
| F       | Contingency                                                           |                                                                        |                                    |                               |                                        |                                         |                                 |        |           |                                                                        |
| G       | SMS                                                                   |                                                                        |                                    |                               |                                        |                                         |                                 |        |           | SMS service charges as applicable will be borne by the user department |
|         | <b>Grand Total (A+B+C+D+E+F)</b>                                      |                                                                        |                                    |                               |                                        |                                         |                                 |        |           | <b>Rounded off to</b>                                                  |

The estimated project cost of Rs. xxxxx /- (Rupees xxxxx only) is for the first year only. If there is any increase in requirement over and above the scope of work given, the cost may require to be revised in tune with revised resources requirement.

For succeeding year of implementation/continuation of the project, one man power indicated in row ‘A’ of above table will be required for which financial provision of Rs. xxxx/- (Rupees xxx only) per annum is to be made. During maintenance/continuation of the project, if there is any additional requirement over and above the scope of work given, the maintenance/continuation of the project cost may require to be revised accordingly.

## **6.1 Hosting**

The hosting environment of the application will be cloud.

(a) Server Specification for Initial Deployment is indicated below.

- 1) DB Server : Linux Ubuntu 22.X , RAM 8 GB , 8 Core VCPU
- 2) App Server : Linux Ubuntu 22.X , RAM 8 GB , 8 Core VCPU
- 3) Additional Storage Space for document storage : 2 TB
- 4) Backup DB and APP Server: No of Linux Ubuntu 22.X, RAM 8 GB, 8Core VCPU.

Depending upon the usage / hit rate, the server requirements and specification may require to be revised.

(b) If user will decide to host the application in NDC, NIC then cost estimate of hosting charge will be furnished separately.

(c) If the application will be hosted in SDC then PR & DW Department will have to contact E&IT Department, Govt. of Odisha.

## **6.2 Payment Term**

In order to initiate the project, Panchayati Raj & Drinking Water Department, Government of Odisha shall make 100% payment of the project cost as indicated above in point no. 7 in advance in form of A/C payee draft / NEFT / RTGS in favour of NICS I payable at New Delhi against the Proforma Invoice (PI) of NICS I which will be placed after obtaining necessary approval from Panchayati Raj & Drinking Water Department, Government of Odisha .

## **7. Roles and Responsibilities**

The responsibilities of Panchayati Raj & Drinking Water Department (PR and DW Department), NIC and NICS I will be as described below.

### **PR & DW Department**

- Nodal Officer of the project is to be identified
- Stakeholders of the application to be identified with reporting hierarchy
- Approval of Project Proposal
- Funding of the Project and timely release of funds
- Acceptance of SRS
- Finalisation of all Input Formats

- Formats of data collection which are to be consumed using external APIs
- Requisition letter to be send to owner department of the applications from which APIs to be consumed.
- API for collection of real time data from the operational applications
- Identification of KPI's based on the data availability through API's.
- Necessary feedback/suggestions during interim demo and execution of the application
- Ensuring that Project Review Committee and other mechanism in place to support the overall efforts in executing the project
- Full ownership of Database & its Security and administration of the DB.
- Data verification for every input and output of data
- Issue of User Acceptance Certificate after testing of the software
- Ownership of Input data and Processed data at all the stages of the project

**NIC:**

- System Study of the existing system
- Preparing Project Proposal and SRS
- Preparing estimated project cost with respect to NICS prices for outsourced manpower which has been included in the Project Proposal
- Design and development of the application and removal of bugs, if any.
- Technical support for Security Audit of the project through NICS empanelled third-party audit.
- Provision of Testing of Software by the users
- Capacity building/training to the users/trainers
- Execution of the project and co-ordination for completion of the project
- Technical support for the project

**NICS:**

- Provide hired manpower support
- Provide Resource (VM/Storage etc. on request)
- Provide NICS empanelled third-party vendor for security audit
- Provide utilization certificate of funds utilised on request.

## 8. NIC Officers – Contact Details

| Sl. No. | Name                    | Designation                                      | E-mail Address  |
|---------|-------------------------|--------------------------------------------------|-----------------|
| 01      | Dr. Ashok Kumar Hota    | Dy. Director General & State Informatics Officer | sio-ori@nic.in  |
| 04      | Shri Rama Krishna Sahoo | Director(IT) / Scientist-E                       | rk.sahoo@nic.in |
| 05      | Ms. Jyoti Gupta         | Scientific Officer/Engineer-SB                   | j.gupta@nic.in  |

## 9. Abbreviation used

| Abbreviation | Description                                |
|--------------|--------------------------------------------|
| APIs         | Application Programming Interfaces         |
| DB           | Database                                   |
| CRUD         | Create, Read, Update, Delete               |
| DAO          | Data Access Object                         |
| E&IT         | Electronics & Information Technology       |
| ICT          | Information Communications Technology      |
| IT           | Information Technology                     |
| MVM          | Model Village Mission                      |
| MAGY         | Mahapurusha Adarsha Gram Yojana            |
| NIC          | National Informatics Centre                |
| NICSI        | National Informatics Centre Services Inc.  |
| PR & DW Dept | Panchayati Raj & Drinking Water Department |
| REST         | REpresentational State Transfer            |
| SDC          | State Data Centre                          |
| SMS          | Short Message Service                      |
| SRS          | Software Requirements Specification        |
| UAT          | User Acceptance Testing                    |

## **10. Terms and condition**

1. Technical Artifact Ownership: IPR or Ownership of the source code of this application would remain with NIC. NIC will hold Intellectual Property Rights to all the artifacts developed/produced under this project and shall have the full right to modify the technical artifacts for use by other Departments at all levels of Government. NIC has rights to claim award(s)/felicitation(s) and submit related technical papers for its intellectual property. User shall have the perpetual right to use the same within the department. For marking any change in the artifacts, user needs to acknowledge the contribution of NIC and obtain the consent of NIC.
2. Manpower rate may vary from time to time as per the NICSI rate.
3. As per advisory released from MeitY, Government of India, TLS/SSL certificate is mandatory for all government websites and procurement of SSL certificate for the application is to be borne additional by the user Department.
4. As per CERT-In guidelines and NIC policy, all internet facing applications are required to be security audited periodically (Once in a year or as and when changes are undertaken in the application) to avoid security incidents and the cost of audit to be borne additional by the user department.
5. Additional Cost & Time: Under special circumstances, if any delayed feedback or modifications to already finalized model or released system are requested, the User Department may have to incur additional cost and time.
6. Data and Business Process Ownership: The Ownership of stored Data, entire transactional data, content data, validations and business rules/ functional requirements belongs to the User department.
7. No Data Access to NIC: NIC will not keep access right of the actual data inserted/generated in the IT Application of User department. The real-time actual implementation and operation of the IT Application will be the prime responsibility of the user department. The User Department shall nominate an officer who has the complete and exclusive access to the stored data. This officer can be trained by the NIC Project Team for this purpose.
8. RTI queries: All RTI queries related to the IT Application are to be answered by the User department. If any technical queries are there, NIC may provide inputs to the Nominated/ designated officer of the department, provided it is technically feasible.
9. Queries from law enforcing, auditing and Regulatory Functionaries: All queries from Government law enforcing, auditing and Regulatory agencies related to the Project IT

Application are to be answered by the User department. NIC shall not be responsible to answer any such queries. The User department should envisage the reports and output required from the IT Application in advance, which will become the part of the IT Application. NIC can help in preparation of exceptional reports in the IT Application to answer such queries by the Nominated/ designated official of the User department.

10. Cleaning of Sample/test Data: The NIC uses sample/test data for IT Application engineering activities, it is the responsibility of the User to clean the sample data before go-live so that there will not be any harm or legal issues due to sample data.
11. Indemnification: NIC should be indemnified from all legal and associated consequences arising due to the usage of the IT application.
12. Sensitization of Govt. Laws. Acts/ Rules and Regulations: User department should give due consideration to sensitize themselves with existing various Government Laws/Acts/ Rules and Regulations such as IT Act 2000, IT Act 2008 (Amendment), IT Rules 2011, Aadhaar Act 2016, Cyber Security Laws, Role of Cert-in, Sensitive Personal Data Information Rules and Privacy of Personal Data etc. issued by Government of India from time to time.
13. Cost towards SMS service charges as applicable will be borne by the user department additionally.

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